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Space definition for match analysis in soccer.

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Abstract

Studies on match analysis in soccer frequently use some kind of geometrical field division in order to situate and classify the actions observed. However, as an invasion game, with both teams acting in the same regular space, intertwining and creating play configurations of outstanding complexity and dynamism, a more appropriate procedure to locate actions encompassing its tactical meaning seems to be in relation to opponent distribution in the field. Aiming to provide a contextual notation of actions, this paper proposes the utilization of a match analysis instrument based on the opposition's spatial configuration called Space of Defensive Occupation (SDO). SDO emerges from the application of the concept of space effectively covered (Merand, 1976; Gréhaigne, 1992) to the entire team while defending. Apart from peripheral limits, the position of players inside the SOD is also considered, resulting in internal subdivisions and generating different zones. For the purpose of studying the relationship between the actions of assistance pass, reception and shot and SDO zones, we have analysed 112 shot attempt situations from seven matches of Brazil and Germany in the 2002 World Cup. Variables were classified into four groups:(1) zone of action's occurrence – frontal (F), central midfield (cM), peripheral midfield (pM), central defence (cD), peripheral defence (pD), central space behind the defence (cB), peripheral space behind the defence (pB);(2) action – assistance pass, reception and shot; (3) shot attempt result – successful or unsuccessful;(4) team – Brazil, Brazil's opponents, Germany and Germany's opponents. Chi-square tests indicated that cD is proportionally the main reception and shooting zone, while cB and pB presented the highest relative efficiency. cM and pM are essentially assistance pass zone, though cM's relative efficiency is specially low. Both of these showed the smaller relative efficiency to reception and shot. For the assistance pass, cD is the most efficient; conversely, there is no significant difference between successful and unsuccessful distributions among SDO zones. According to the spatial distribution of actions related to successful shot attempts, assistance pass was most diversified and unpredictable (J=0.94) than reception (J=0.83) and shot (J=0.76). Brazil played a larger proportion of assistance passes in central zones with reception in central zones than Germany, and a smaller proportion of assistance passes in peripheral zones with reception in central zones than its opponents. Germany did not present significant differences compared to its opponents. The instrument's reliability was tested with values ranging from K=0.73 to 0.90 for inter-observers test and from K= 0.83 to 0.93 for intra-observers test. The analysis of results suggests that SDO is a reliable instrument, capable of spatially characterizing actions so as to point out differences in team performance.

Keywords: soccer, match analysis, opposition, space of defensive occupation, zones, shot attempt situations.

1.0. Introduction.

Notational analysis is based on the choice, classification, observation and record of variables of interest concerning actions which take place during the game. In terms of methodology, it refers to extracting certain happenings from a complex, dynamic and situational whole, as is a soccer game, and grouping them together into different categories, in order to interpret the behaviour of teams or players. Apart from the undisputed contribution of such activity for the understanding of tactical, technical and strategic aspects of the game, (Hughes, 1993; Garganta, 2001), the reduction in complexity of the data gathering process inevitably leads, on a higher or lower scale, to some loss in the situational significance of the actions. In this respect, for a complete characterization of the game action, Garganta (1997) considers it necessary to encompass the different dimensions of performance within the game, present in every action – task, time, space.

In several studies, the spatial dimension was addressed via geometrical divisions of the field. A great number of field divisions can be observed in different works (Castelo, 1999; Garganta et al., 1995; Hughes et al., 1988; Mombaerts, 1996; Yamanaka et al., 1993). This procedure, closely related to the formal space of the game, does not seem coherent with the spatial characteristics of the game, essentially dynamic and informal. As soccer is an invasion game, the constant dynamic interaction between the teams, resulting from momentary circumstances of the rapport of strength causes continuous modifications in the distribution of players in the field, giving it a fluid and dynamic character, which expresses itself visibly and objectively in the creation and transformation of spaces. From a dynamic systems perspective, it refers to configurations of play and their evolution, which involve, respectively, structural aspects - mainly the spatial organization of the players and their respective teams - and functional aspects - related to transformations brought about by the opposition relationship along time (Gréhaigne and Bouthier, 1995; Gréhaigne et al., 1997).

Methodologies based on the study of evolution of configurations of play seem to provide a very rich understanding of the spatial and time dimensions of actions, as every action is considered within its own situational context and within a time line of events. However, notational analysis, which is based on accumulation and treatment of data, requires spatial modelling of the game, one which takes into account the actions in their spatial and time dimensions, and, additionally, offers the possibility of classifying them into general spatial categories. Doing that, within criteria coherent with the informal and dynamic spatiality of the game, would enable the analysis of the spatial characteristics of different actions, sequence of actions, of teams and players. This paper aims at proposing a system of observation apt to guide such classification.

The reference criterion used to locate actions has been the momentary distribution of the opponent team in the field, in the defence phase. The spaces generated by such distribution constitute the system's spatial characteristics. Our hypothesis is that a spatial modelling of the game for notational analysis based on the organizational

structure of the opponent team, according to the positioning of its defence system, shall enable the classification and characterization of the actions, offering a general idea of the situational context in which they took place. Additionally, it would be possible to examine the dynamic aspect of the game by analysing actions as regards the time sequence in which they took place. After describing the conceptual construction and examining the reliability of the observation system, we shall present an initial characterization using the spatial analysis of actions of assistance, reception and shot, taken from shot attempt situations of the World Cup of 2002.

2.0. Method.

2.1. The Observation system

In order to build the observation system, the notion of effective space covered (ESC) was adopted (Mérand, 1976) used in the works of Gréhaigne (1992) and Garganta (1997). The ESC is determined by the successive position of players situated within the perimeter of the group of players of both teams at a given moment T, generating a polygonal surface figure – goalkeepers are not taken into consideration (Fig 1). The ESC enables locating actions in the periphery – beside, behind and opposite – or in their interior (Gréhaigne, 2001).

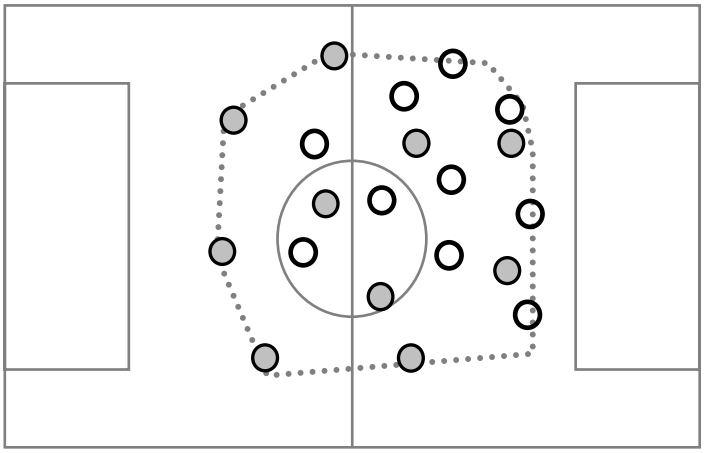

Fig 1: Effective Space Covered

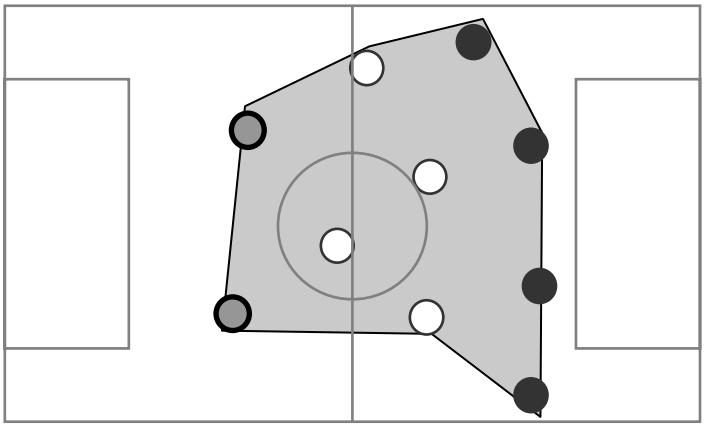

Fig 2: Example of Space of Defensive Occupation. In black: Defence sector; in white, Midfield sector; in grey and black, Offensive sector.

Application of the (ESC) to the team in the defensive phase generates a polygonal figure representing the Space of Defensive Occupation (SDO). Like ESC, SDO also enables locating actions in their periphery or their interior, albeit in greater detail via definition of defence, mid field and attack sectors, which originate the different zones of the SDO (Fig 2).

2.2. Criteria for the definition of SDO Sectors.

Defence, mid field and attack sectors are formed by players individually classified as per their position, according to two criteria: (1) centrality/periphery as regards the SDO and (2) coverage from team member as regards position of the ball in the defensive action. To this end:

1. Defence player is the one positioned in the periphery of the SDO, who receives coverage from a player also in the periphery;
2. Mid fielder is a player who is (a) within the SDO – whether or not receiving coverage from a defence player, or (b) in the periphery of the SDO, from the moment of receiving coverage from a player within the SDO;
3. Attacking player is the one who is in the periphery of the SDO, who does not receive coverage from players of the defensive sector.

2.3. Definition of the SDO Zones.

The spaces between the sectors defined above originate the SDO zones.
The space "behind the defence", between the defence sector and the back line originate zone "B"; the space "opposite the defence", between defence and mid field sectors originate zone "D"; the space between mid field and attack originate zone "M"; the space opposite the attack sector originates zone "F".
Each of these zone can be subdivided into central and peripheral ones, resulting in zones "cB", "pB", "cD", "pD", "cM", "cM", "cF" and "pF" (Fig 3).

In order to differentiate central from peripheral zones, we have established the centrality/periphery criterion applied to that sector. Thus, in case the action takes place between a player from the periphery of the sector and the closest side line, it is a peripheral zone. In case it happens between players of one sector, it is a central zone.

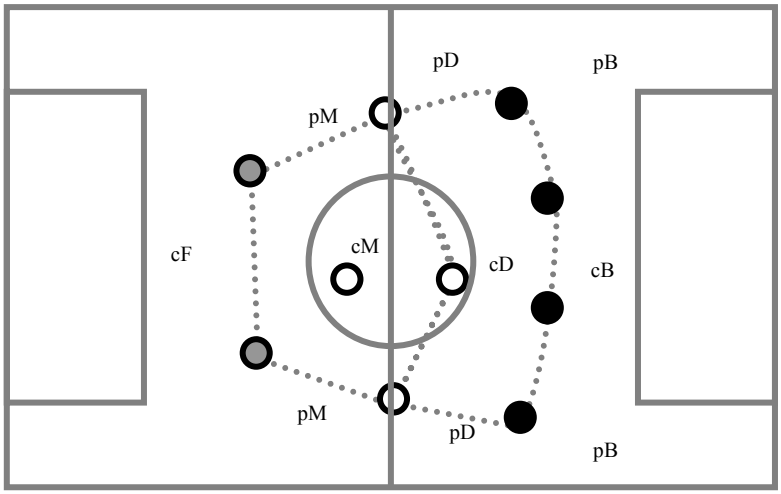

Fig. 3. SDO zones

2.4. Reliability.

Two experienced observers in notational analysis were given a description and criteria of the system one week before the test, in order to study it. A video recording of the images broadcast by Spanish television of the first half of the game Barcelona vs Real Madrid of the 2004-2005 season of the Spanish Soccer League was used. Fifty two situations of pass-reception in which the Barcelona team executed the reception action in zones cD, pD, cB, pB were selected for analysis. The observers indicated the relative location (SDO zone) of each pass and reception. Reception actions were not selected for their technical expression, i.e. receive and control the ball, but rather, for the tactical sense of receiving a pass – thus, passes and first hand kicks and header deviations were also considered reception. Actions which presented visual difficulties as for the position of players (very closed image angle, or field level imaging) were disregarded.

For the purpose of standardization, both for passes and reception, the observers were instructed to pause the video at the exact moment when a player comes into contact with the ball, for then to consider the SDO configuration of the Real Madrid team and point out the location of the action. An interval of three weeks between testing and re-testing was used, so as to prevent memorization.

Many receptions take place within zone borders, casting doubt as to location. We have established that, in the event that an observer is unable to determine the SDO zone, the location of the next action over the ball, representing the immediate consequence of the reception action, should be pointed out.

At times the dynamics of the defence sector makes it hard to distinguish reception in zone D from reception in zone M. These are situations in which the defence sector changes its configuration in the gap between one pass and the next reception: usually a defence player moves forward to directly oppose the attacker who will receive the ball, while the others are either stopped, get near, or move back to ensure more depth

and coverage to the defence. In such situations, the player remained seen as a defence player, even if he is within the SDO, if he still has coverage from the other defence players. Consequently, in order to select actions of reception, it was necessary to pause the video in two moments (execution of pass and reception), and observe the evolution of the distribution of defence players in these two moments. This flexibility of the model allows us to treat actions in a continued manner, not isolating them completely from the preceding action, enabling the analysis of both the structure and functionality of defence.

Kappa statistics (Landis and Koch, 1977) for categorical data was used in order to measure the level of agreement intra and inter observers for the variables pass and reception, under the following formulae:

$k=(\pi o - \pi e)/(1-\pi e)$
πo = probability of agreement observed
πe = hypothetical expected probability

Values of k between 0.61 and 0.80 are considered strong and between 0.81 and 1.00 nearly perfect.

Table 1. Kappa statistics for agreement test

	AvsB Test	AvsB Retest	AvsA	BvsB
Pass	0.85	0.72	0.83	0.85
Reception	0.86	0.75	0.90	0.93

2.5. Analysis of Shot attempt situations.

For an initial characterization of the SDO we studied its relation with assist, reception and shot actions in 112 shot situations of game in progress of the World Cup of 2002.For the attack to be able to take a shot, it means that enough space-time advantage over defence was conquered to attempt scoring. That is why these actions were chosen. Therefore, it would be possible to see how these offensive actions behave in relation to the SDO.

The seven eliminating games of the finalists - Brazil and Germany - were selected for analysis. From a total 187 shots, we used only the ones that were preceded of assistance from a team member, excluded situations of replacement of ball in the game (Table 2)

Variables were classified into four groups: (1) zone of action's occurrence – frontal (F), central midfield (cM), peripheral midfield (pM), central defence (cD), peripheral defence (pD), central space behind the defence (cB), peripheral space behind the defence (pB);(2) action – assistance pass, reception and shot; (3) shot attempt result – successful or unsuccessful; (4) team – Brazil, Brazil's opponents (BOP), Germany and Germany's opponents (GOP).

Table 2. Action before shot attempt

Assist				Without assist				
Pass	Free kick	Corner	Throw in	Rebound	Interception	Tackle	Rebound free kick	Free kick
112	14	9	2	19	4	1	9	17
137				50				
187								

For the purposes of analysing action distribution within zones of the SDO, data of zones cF, pF, cB and pB were grouped into zones F and B, in view of the small number of actions within these zones.

Successful actions were those in which shot resulted in goal, goalkeeper's defence or ball against cross bars. Unsuccessful actions were those that generated all other possible results of shot attempts, among which, the most common were: shot wide, deviated or blocked shot by the defence.

In order to analyse the behaviour of teams based on the spatial distribution of actions, we first effected a zone grouping, ensuring sufficient samples to form contingency tables and avoiding expected frequencies below 1. The zones were grouped into peripheral and central - zones pF and cF and pB were considered peripheral (PZ) together with zones pM and pD, the others were considered central (CZ). Afterwards, in order to comparatively analyse the final movement of teams prior to finalization, the actions of assist and reception were taken in sequence, together with the relevant spatial categories. In other words, each assist zone was connected to the reception zone which ensued. Thus, the following assist-reception categories emerged: CZ – CZ, CZ – PZ, PZ – CZ, PZ – PZ. It is worth mentioning that category PZ – PZ encompasses peripheral actions of assistance – ipsi lateral receptions and counter lateral receptions in relation to the SDO.

The teams, divided into four groups as shown above, accumulated the data of all their games. This way, the data of the teams of Brazil and of Germany correspond to the sum of the data in each of the four confrontations, from the eighths finals to the final game. The BOP and GOP data correspond to the total sum of data of different teams that confronted Brazil and Germany, respectively. As a consequence, the data of Brazil and GOP cannot be compared, since the data of Brazil originated from the confrontation with Germany are present in the two groups. The same applies to comparing Germany and BOP. Thus, only three comparisons were made: Brazil and Germany, Brazil and BOP, Germany and GOP.

2.6. Statistical Treatment.

Chi-square (*x2)* independence tests were used to test the association between the action variables and SDO zones, shot result and SDO zones under action, team and grouped SDO zones under action, and between team and grouped SDO zones under actions of assist and reception in sequence. Shannon's rate of corrected diversity was calculated to measure the degree of spatial unpredictability of each action.

The diversity index enables measurement of diversity or unpredictability of a distribution (Zar, 1984). In other words, the more homogenous the distribution, the higher the uncertainty of behaviour of the variable studied. Below, the formulae used for calculating Shannon's diversity index (H), and for the corrected diversity index, which is the relative diversity index (J):

$$H = -\sum_{i=1}^{k} \pi i \log \pi i \qquad J = H/Hmax; \text{ where } Hmax = \log k$$

πi=proportion of each category; k = number of categories

Chi-Square (*x2)* goodness of fit test was used in two different cases. First, to test the association between the actions and SDO zones. . Later to test the association between the grouped zones of the SDO and actions of assist and reception in sequence, in order to examine the trends of circulation of the ball in these actions in relation to the SDO.

3.0. Results.

Table 3. Spatial distribution of actions among SDO zones

Actions	SDO zones						Total
	B	CD	pD	cM	pM	F	
Assist	6	20	27	25	29	5	112
Proportion	0.05	0.18	0.24	0.22	0.26	0.04	1.00
Reception	12	47	26	16	11	0	112
Proportion	0.11	0.42	0.23	0.14	0.10	0.00	1.00
Shot	14	54	18	16	10	0	112
Proportion	0.13	0.48	0.16	0.14	0.09	0.00	1.00
Total	32	121	71	57	50	5	336
Proportion	0.10	0.36	0.21	0.17	0.15	0.01	1.00

Table 4. Results of X2 goodness of fit test on actions distribution among SDO zones

	Greater proportion	Smaller Proportion	p value
Assist	pM. pD	B. F	<0.001
Reception	CD	pM. B	<0.001
Shot	CD	pM. B	<0.001

Table 6. Ball circulation between successive actions

Circulation type	Successive actions	
	Assist – Reception	Reception – Shot
Same zone	26	93
Proportion	0.23	0.83
Different zone	86	19
Proportion	0.77	0.17

Table 7. Spatial distribution of Assists according to shot's result and Chi Square Independence test value.

Result	SDO zones							*x2*
	B	cD	PD	cM	pM	F	Total	
Successful	3	11	9	6	13	0	42	8,82
Unsuccessful	3	9	18	19	16	5	70	
Total	6	20	27	25	29	5	112	
Success proportion	0.50	0.55	0.33	0.24	0.45	0.00	0.38	
								Ns

Table 8. Spatial distributions of Reception according to shot's result and Chi Square Independence test value.

Result	SDO zones						*x2*
	B	cD	PD	cM	pM	Total	
Successful	8	18	11	4	1	42	9.48
Unsuccessful	4	29	15	12	10	70	
Total	12	47	26	16	11	112	
Success proportion	0.67	0.38	0.42	0.25	0.09	0.38	
							P<0.1

Table 9. Spatial distribution of Shot according to shot's result and Chi Square Independence test value.

Result	SDO zones						*x2*
	B	CD	PD	cM	pM	Total	
Successful	9	22	8	1	2	42	12.87
Unsuccessful	5	32	10	15	8	70	
Total	14	54	18	16	10	112	
Success proportion	0.64	0.41	0.44	0.06	0.20	0.38	
							P<0.05

Table 10. Distribution of successful actions and diversity index.

Actions	SDO Zones					Total	J
	B	cD	PD	cM	pM		
Assist	3	11	9	6	13	42	0.94
Reception	8	18	11	4	1	42	0.83
Shot	9	22	8	1	2	42	0.76
Total	20	51	28	11	16	126	0.91

J= Shannon diversity index

Table 11. Spatial distribution of teams assist and *x2* independence test value.*

	Assist Zone			*x2*
	Central	Peripheral	Total	
Brazil	25	18	43	5.13
BOP	11	23	34	
Total	36	41	77	
				P<0.05
Germany	8	20	28	1.36
GOP	12	14	26	
Total	20	34	54	
				ns
Brazil	25	18	43	6.04
Germany	8	20	28	
Total	33	38	71	
* With Yates correction				P<0.05

Table 12. Spatial distribution of teams receptions and *x2* independence test value. *

	Reception Zone			*x2*
	Central	Peripheral	Total	
Brazil	26	17	43	2.31
BOP	26	8	34	
Total	52	25	77	
				ns
Germany	16	12	28	0.47
GOP	17	9	26	
Total	33	21	54	
				ns
Brazil	26	17	43	0.13
Germany	16	12	28	
Total	42	29	71	
* With Yates correction				ns

Table 13. Goodness of fit test to Assist – Reception zone

	Assist - Reception Zone				
	CZ - CZ	CZ- PZ	PZ- CZ	PZ- PZ	Total
O	27	21	41	23	112
E	28	28	28	28	112
x2	0.0	1.8	6.0	0.9	8.7
					P<0.05

Table 14. The *X2* Independence test to compare teams assist – reception zones.

Team	Assist - Reception Zone														
	CZ- CZ			CZ – PZ			PZ - CZ			PZ - PZ			TOTAL		
	O	E	*x2*	O	E	*x2*	O	E	*x2*	O	E	*x2*	O	E	*x2*
Brazil	16	12,8	0.78	8	7.3	1.03	11	16.2	1.67	8	6.7	0.01	43	43	3.49
BOP	7	10,2	0.98	5	5.7	0.10	18	12.8	2.11	4	5.3	0.32	34	34	3.50
Total	23	23	1.76	13	13	1.13	29	29	3.77	12	12	0.33	77	77	6.99
															p<0.1
Germany	3	6,2	1.67	6	4.7	0.38	12	10.4	0.26	7	6.7	0.01	28	28	2.32
GOP	9	5,8	1.80	3	4.3	0.41	8	9.6	0.28	6	6.3	0.01	26	26	2.49
total	12	12	3.47	9	9	0.79	20	20	0.53	13	13	0.02	54	54	4.81
															ns
Brazil	16	11,6	1.66	8	8.6	0.04	11	14.1	0.66	8	9.2	0.15	43	43	2.51
Germany	3	7,4	2.61	6	5.4	0.06	12	8.9	1.04	7	5.8	0.23	28	28	3.94
Total	19	19	4.27	14	14	0.09	23	23	1.71	15	15	0.38	71	71	6.45

O= observed frequencies; E= expected frequencies;*X2*= Chi Square values p<0.1

4.0. Discussion.

4.1. Reliability.

Reception in zones D and B were taken as criterion for selection of actions of pass used in the test. Passes and related receptions were subjected to the observer's analysis. In principle, any offensive action, with or without ball, could have been used. Pass and reception actions were selected because they would be used in the analysis of situations of finalization carried out afterwards.

The results of the experiment, as depicted in table 1, show a higher concordance level for reception than for passes. Obviously, as we restricted reception zones to only four, the probability of disagreement becomes smaller than for passes.
In spite of the high values, in no test was agreement perfect, evidencing that the criteria for determination of SDO zones are general guidelines to lead the observation, and that they certainly do not cover all the diversity of situations found in soccer game. Besides, other factors contribute to a disagreement margin. Visualization factors – very often the limits between zones is not clear due to broadness and angle of televised broadcasts – and interpretation factors – not only of criteria for definition of zones, as, probably the very situations of the game – can have great influence. One cannot expect an experienced observer to be a mere carrier of a fixed structure. On the contrary, it seems more plausible to admit that he embraces the criteria and adapts them to the game situations according to his own perception of the situation, taking the position of agent over the SDO. This might account for the fact that most intra observer levels of agreement were higher than inter observer agreement levels. This hints at the idea that the agent fills the gaps of the criteria, making use of his own referential. As a consequence, the objectivity of this system could and should be improved.

The use of televised images presents the greatest limitation to this study, as very often the filming does not encompass all the players of the defence sector involved. In order to minimize this factor, we allowed observers to forward and rewind the film as they deemed fit and as many times as they wanted, in order for a mental picture of the situation to be created whenever necessary.

4.2. Shot attempt situations.

Table 3 presents distribution data and proportions of actions in the SDO zones. In the last line of the table we can se that a little more than half the shot attempt situations are concentrated in zones D. Zone cD alone represents a little over one third of the total number of actions. Zones B are present in only 10% of the actions and zones F in only 1%, without reception and finalization. Zones M are present in nearly one third of the actions.

Assists are greatly distributed among zones cD, cM, pD, pM, with the two latter having the highest number of occurrences. Zones B and F presented the lowest proportions. Receptions and shots, on the other hand, presented very similar trends. Both are largely concentrated on zone cD, with few occurrences in zones pM and B.

In table 4, chi-square goodness of fit tests carried out for each action showed that these differences in the distribution of actions among the zones, seen in table 3, are significant for all actions ($p<0.001$), meaning that the actions respond differently to SDO locations. These differences in spatial behaviour of assists and receptions or shots, as well as the similarities between receptions and shots, confirmed with the chi-square independency tests in table 5 derive from the very dynamic of the finalization situation. Table 6 clearly shows that, in most cases, ball circulation between assist and reception takes place by changing SDO zones. This explains why these actions have different spatial characteristics. On the other hand, most times, the reverse happens between reception and finalization: there is no SDO change, making the spatial characteristics very similar.

The inexistence of marked spatial differences between actions of reception and shot demonstrates that there hardly ever is a change in the location of the attacker in relation to the SDO from the moment of reception to the moment of shot. It is known that the high pressure space/time in conditions of finalization (Harrys and Reilly, 1988) reduces the possibilities of the finalizing player to lead the ball, forcing him to shoot quickly, in most cases with maximum two contacts (Olsen, 1988; Castelo, 1999). Besides, this important conclusion demonstrates, as in other works, (Gréhaigne, 1992; Gréhaigne et al., 1997; Hughes, 1998) that the overtaking, unbalance or perturbation of the defence system, in situations of game in progress resulting in shot, rarely happen as a result of action with ball on the part of the shooting player. It is created previously.

On the other hand, spatial differences among actions of assist and reception demonstrate that certain actions take place preferably in certain zones. Besides, the efficiency of actions varies according to the zone in which they happen. This differentiated distribution and efficiency of actions in zones simultaneously result in different technical-tactical profiles for the zones, and spatial profiles for the actions.

The success or failure of an attack action seems not to be associated to the zones of the SDO as regards assist, in a statistically significant manner. In spite of that, we observe a mean difference of 31% among zones B, cD and pM, of higher efficiency (success proportion), and zones pD, cM and F of lower efficiency (table 7).

As for reception, the test indicates the existence of association among zones of the SDO and the result of the action ($p<0.1$). Zones B and pM are the ones that most distinguish themselves, contributing for the result of the test in a significant way. The former, for the high percentage of success, and the latter, for the high percentage of failure (Table 8). It is worth mentioning that although zone cD represents mean percentage of success, it is also responsible for the higher number of receptions in successful finalizations.

In shots, the association is even stronger ($p<0.05$). The zones that most contributed to this difference were zone B, owing to the high percentage of success, and zone cM, for the high proportion of failure. Again, zone cD carries the highest number of successful shots, even with a mean proportion of success (table 9). Zone F was not taken into account for reception and finalization, as it id not present any observation (Table 3).

The values presented in table 10 indicate that assist is the most spatially unpredictable of actions regarding success. Successful shot situations do not depend on spatial condition of the SDO that is specific to assist. This aspect is of paramount importance, as it evidences the unpredictable and variable character of opportunity creation in the game. The only zones that seem to be less related to assist are zones B and F, although zone B depicts efficiency levels of 50%, while zone F does not present any successful shot attempt in the five assists observed.

Actions of reception and shot, with clear concentration around zone cD, on the other hand, present lower levels of diversity, stressing the importance of this zone for the success of the finalization. Zone B, for efficiency (64% reception and 67% finalization) and zone pD, for the number of receptions, appear in a secondary position.

By and large, these results show that the importance of zones in reception and shot seems to grow as the action approaches the opponent's defensive sector or its overtaking – possibly due to greater nearness to the target and smaller number of defence players obstructing the attacking action. In this respect, zone F has practically no importance whatsoever, zone M has little importance, zone D is very important, and zone B, in spite of lower frequency, is very important in efficiency.

In short, we can describe the characteristics of each zone according to the results presented as follows:

In zone cD we find the highest number of reception and shots, the highest number of receptions in successful shots, and high number and proportion of successful assists. These observations evidence the fundamental importance of zone cD in shot attempt situations.

Conversely, zone pM accounts for 15% of the total number of actions, high proportion of assists of average efficiency (45% of successful shots), and few receptions and shots– with extremely low efficiency.

Zone cM accounts for 17% of the actions. It stands out for low efficiency in shot, although the proportion of success for assist and reception is likewise low.

Zones B, in spite of the low number of actions, stand out for high efficiency of assist and specially of reception and shot actions. The 112 actions used as sample in this paper are, undoubtedly, insufficient for accurate characterization of zones B. Grouping of zones cB and pB meant that some important information was lost. As reception in zone B represents physical overcoming of the SDO barrier – as, under this condition, there are no defenders between the attacker with the ball and the goal or goal line – it is hardly consented by the defence; and when it is, quick efforts of defensive recovery ensues. The Brazilian team, for example, did not incur any finalization derived from reception in zones B in the four games observed.

Of all the zones studied, zone pD is the one that shows a more even distribution and mean values of efficiency among the three actions.

Only actions of assist and reception were used in team comparison, as the Chi Square tests did not show association between the variables reception and shot and SDO zones, evidencing the similarity among distributions and their frequency.
Table 11 shows assist distribution of the teams with the result of the Chi Square test. When compared to its opponents and to Germany, Brazil demonstrated a significantly higher proportion of central assists.

As for receptions (table 12), although Brazil clearly presents higher proportion of peripheral receptions than its opponents, there was no statistically significant difference. (17/43 vs 12/28).

Table 13 shows the goodness of fit test for distribution of the 112 actions within the four categories of assist-reception. It is possible to notice higher concentration of observations in category PZ – CZ as the main factor of heterogeneity of distribution greatly contributing for the value of the Chi Square test. In secondary position smaller number of actions CZ – PZ appears.

According to table 14, Brazil greatly differentiated from its opponents ($p<0,1$) mainly due to smaller proportion of category PZ – CZ (assist in peripheral zone with reception in central zone). The main difference evidenced in the comparison between Germany and its opponents is, likewise, in category CZ – CZ owing to the low proportion achieved by Germany – as values *X2* of other categories did not present major discrepancies, the independence test did not reach statistical relevance.

Brazil and Germany distinguished themselves first for category CZ – CZ, in which Brazil presents a much higher proportion than Germany. Second, for category PZ – CZ, in which Germany presents a much higher proportion than Brazil.

These results show that the tactical behaviour of high level teams in shot attempt situations may vary within the same team, and also, from team to team. Such preferences in creation of situations of finalization seem to be related to the style of play of the teams, and, not necessarily, to parameters of efficiency. Besides, comparisons between Brazil and BOP and Germany GOP enable visualizing not only the form of attacking the opponent, but also the form of allowing attack, which supplies useful information for the tactical defensive analysis of the team.

Grouping central and peripheral zones certainly hinders a more detailed understanding of teams' behaviour. However, that was necessary to avoid a large number of assist – reception categories, as well as to ensure a sufficient number of observations in each of them.

We observe that isolated analysis of assists and receptions revealed that only Brazil seems to prefer assists in central zones, while BOP and Germany seem to favour peripheral zones, with no significant differences regarding reception. On the other hand, analysis of actions in sequence (i.e. zone of assist – zone of reception) point towards Brazil's preference for central assists with central receptions, as opposed to peripheral assists with central receptions. This difference, not minor in the analysis of result, stresses the importance of considering actions in sequence within the time dimension of the game. Being a dynamic system, it is not enough to know the condition of the system in one moment to know its past, present and future. The

unpredictability of dynamic systems in phase transitions with forking pints, require knowledge of their evolution. The time aspect of the analysis cannot be overlooked, if we intend to understand the transformations suffered by the system (Prigogine, 1996).

4.3. Possible applications.

Once the position of an attacking action in the SDO represents the degree of invasion or even the degree of overtaking of the defensive block of the opponent, it is related to the distance and to the possibility of shooting against the opponent's target – as we observed, the efficiency of assist, and number and efficiency of reception and shot increases in more advanced zones of the SDO. Considering both the notion of distance and the degree of defensive interference, the SDO could constitute an order parameter in the systemic requirements discussed by McGarry et al (2002). In this respect, the position of the ball in attacking actions in more distant zones, such as zones F and M, would mean condition of stability of the game system, demonstrating balance between attack and defence. Zone cD could represent condition of instability in which the defence is at risk if the ball holder has enough time and space, and if the target is close, a shot can occur with high possibility of success. If the target is far, or if the space and time conditions are not appropriate for shooting, a penetrating pass, or even an assist may ensue. Zones B would represent instability condition of the system with unbalance favouring attack given its high efficiency. Therefore, having ball control in one of these zones could possibly be accepted as an indicator of success of ball possession alternative to reaching the attacking ⅓ of the field.

The relevance of this and other possible applications of the SDO depend upon a broader spatial characterization of actions, based on bigger volume and diversity of game sequences. Moreover, the control parameters also should be identified via study of indicators of the time relation of actions that can measure the speed and rhythm of the game in each zone of the SDO, and of transition among zones. In this respect, it does not seem odd that actions involving confrontation with the defensive sector take place under higher time/space pressure, demanding more speed in perception, decision making and execution, synchronization and connected sequence of actions.

5.0. Conclusions.

Although there is evidence that there are gaps in its formulation, making room for some subjectivity during the observation, the SDO showed to be reliable.

Action classification in the different zones of the SDO enabled the characterization of assists, receptions and shots spatially. The results evidence differences in distribution and efficiency of actions among the zones of the SDO: assists are evenly distributed in zones pM, cM, pD and cD, with zones cD, B and pM as the ones of highest efficiency. Receptions and shots concentrate mainly on zone cD, with zones B evidencing highest efficiency. More than that, they show that efficiency of shot is higher when these actions happen in zones of confrontation with the defensive sector, demonstrating the importance of the last actions of a shot attempt sequence to take place in these zones.

The spatial classification of assist-reception actions, taken in sequence, was apt to discriminate differences in tactical behaviour of teams, and is more informative than the analysis of isolated actions, albeit limited to two actions. This highlights the value of the time dimension for notational analysis. Still in the matter of spatial dynamics of actions, it was observed that assist and reception nearly always take place in different zones, while reception and shot nearly always take place in the same zone. This is definitely an aspect that requires more in-depth studies involving longer connected sequence of actions, so as to look into the ball circulation *vis a vis* the SDO in different types of sequences. Overall, longer and more diversified sequences, as well as a bigger number of actions, require further investigation to enable a more thorough characterization of the SDO.

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